RESULTS — TMUA 2017 P1

18 / 20

20 attempted, 0 left blank · total time on paper 1:16:40

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TMUA SCORE

9.0 2024+ converted 9.0 Official

Official: TMUA 2017's published score conversion. 2024+ converted: our estimate of the equivalent on today's scale.

BY TOPIC

Circle Geometry	2 / 2
Differentiation	2 / 3
Exponentials and Logar…	2 / 2
General Algebra	1 / 1
General Trigonometry	1 / 2
Geometry	2 / 3
Inequalities	3 / 3
Integration	4 / 4
Polynomial Expansions	1 / 1
Remainder Theorem	1 / 1
Sequences and Series	3 / 3
Transformation of Grap…	3 / 3
Trapezium Rule	1 / 1

BY DIFFICULTY

3.3	1 / 1
4.1	1 / 1
4.3	1 / 1
4.4	1 / 1
4.5	1 / 1
4.7	1 / 1
5.0	1 / 1
5.1	1 / 1
5.2	3 / 3
5.3	1 / 1
5.4	0 / 1
5.6	1 / 1
5.7	1 / 1
5.8	1 / 1
5.9	1 / 1
6.4	1 / 1
6.8	1 / 1
7.4	0 / 1

Per-question breakdown

#	YOUR ANSWER	CORRECT	RESULT	TIME		
1	C	C	✓	2:27	Bookmark	Review
2	C	C	✓	4:42	Bookmark	Review
3	A	A	✓	5:03	Bookmark	Review
4	B	B	✓	1:12	Bookmark	Review
5	C	C	✓	1:34	Bookmark	Review
6	B	B	✓	3:54	Bookmark	Review
7	B	B	✓	3:31	Bookmark	Review
8	A	A	✓	2:46	Bookmark	Review
9	F	F	✓	2:26	Bookmark	Review
10	E	E	✓	10:55	Bookmark	Review
11	A	A	✓	2:46	Bookmark	Review
12	B	B	✓	2:23	Bookmark	Review
13	C	C	✓	2:33	Bookmark	Review
14	F	F	✓	2:29	Bookmark	Review
15	B	B	✓	3:50	Bookmark	Review
16	B	E	✗	3:38	Bookmark	Review
17	D	D	✓	5:23	Bookmark	Review
18	A	A	✓	2:06	Bookmark	Review
19	D	D	✓	8:39	Bookmarked	Review
20	C	E	✗	4:23	Bookmarked	Review

QUESTION

The functions f and g are given by $f(x) = 3x^2 + 12x + 4$ and $g(x) = x^3 + 6x^2 + 9x - 8$.

What is the complete set of values of x for which one of the functions is increasing and the other decreasing?

A $x \geq -1$

B $x \leq -1$ YOUR CHOICE

C $-3 \leq x \leq -2, x \geq -1$

D $x \leq -2, x \geq -1$

E $x \leq -3, -2 \leq x \leq -1$ CORRECT

F $x \leq -3, x \geq -2$

G $-2 \leq x \leq -1$

SOLUTION

The correct solution is option E. For details see UAT official worked answers

Differentiation Difficulty 5.4

20 C E ✕ 4:23 Bookmarked Hide

QUESTION

The lengths of the sides QR, RP and PQ in triangle PQR are a, $a+d$ and $a+2d$ respectively, where a and d are positive and such that $3d > 2a$.

What is the full range, in degrees, of possible values for angle PRQ?

A $0 < \angle PRQ < 60$

B $0 < \angle PRQ < 120$

C $60 < \angle PRQ < 120$ YOUR CHOICE

D $60 < \angle PRQ < 180$

E $120 < \angle PRQ < 180$ CORRECT

SOLUTION

The correct solution is option E. For details see UAT official worked answers

General Trigonometry Geometry Difficulty 7.4

Spent a long time:

10	E	E	✓	10:55	Bookmark	Hide

QUESTION

A curve C has equation $y = f(x)$ where

$$f(x) = p^3 - 6p^2x + 3px^2 - x^3$$

and p is real.

The gradient of the normal to the curve C at the point where $x = -1$ is M.

What is the greatest possible value of M as p varies?

A	$-\frac{3}{2}$	B	$-\frac{2}{3}$
C	$-\frac{1}{2}$	D	$\frac{1}{4}$
E	$\frac{2}{3}$ CORRECT	F	$\frac{3}{2}$

SOLUTION

The correct solution is option E. For details see UAT official worked answers

Differentiation Difficulty 6.4

19 D D ✓ 8:39 Bookmarked Hide

QUESTION

The set of solutions to the inequality $x^2 + bx + c < 0$ is the interval $p < x < q$, where b, c, p and q are real constants with $c < 0$.

In terms of p, q and c, what is the set of solutions to the inequality $x^2 + bcx + c^3 < 0$?

A $\frac{p}{c} < x < \frac{q}{c}$

B $\frac{q}{c} < x < \frac{p}{c}$

C $pc < x < qc$

D $qc < x < pc$ CORRECT

E $pc^2 < x < qc^2$

F $qc^2 < x < pc^2$

SOLUTION

The correct solution is option D. For details see UAT official worked answers

Inequalities General Algebra Difficulty 6.8